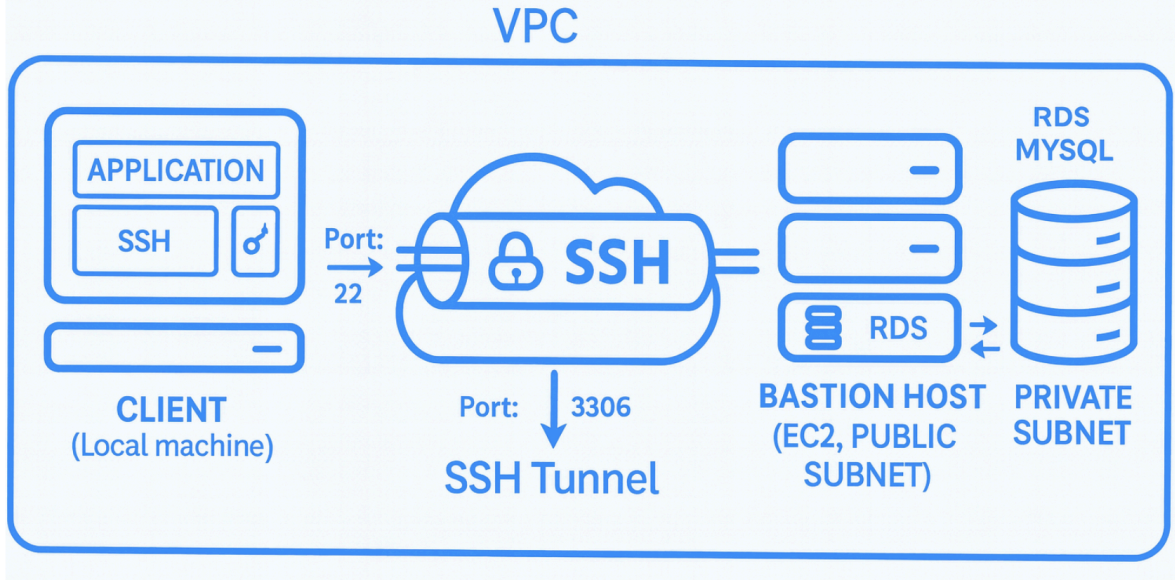


SSH Tunnel ile MySQL Portuna Güvenli Eriřim

SECURE MYSQL ACCESS VIA SSH TUNNEL (BASTION HOST)



SSH Tunnel Nedir?

Modern bulut ortamlarında veritabanlarını doğrudan internete açmak güvenlik açısından büyük risk taşır. Bu yüzden **database sunucuları** genellikle **private subnet** içinde konumlandırılır. Ancak bu durumda dışarıdan bağlantı kurmak zorlaşır. İşte burada **SSH Tunnel** devreye girer.

SSH Tunnel (Port Forwarding), bir bastion host üzerinden güvenli bir “tünel” açarak private subnet içindeki kaynaklara erişmemizi sağlar. Yani public IP’si olmayan bir veritabanına, bastion host üzerinden şifrelenmiş bağlantı kurarız.

✓ Avantajları:

- Public IP açmadan erişim sağlar.
- Trafik şifrelenir, güvenlik artar.
- Minimum attack surface (saldırı yüzeyi) oluşturur.

Bu lab çalışmamızda, bir bastion host üzerinden **RDS MySQL veritabanına** SSH Tunnel kuracağız ve hem **DBeaver** gibi GUI araçlarıyla hem de terminal üzerinden bağlantı sağlayacağız.

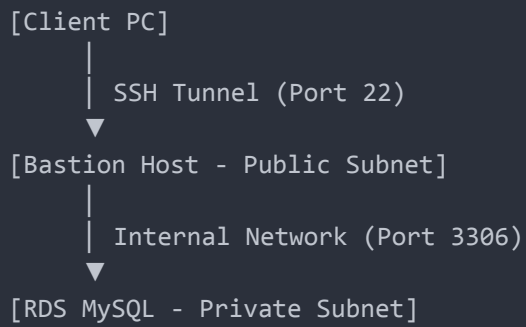
Amaç

- Bastion host üzerinden Private Subnet'teki MySQL'e güvenli erişim sağlamak.
- Database'i public networke açmadan bağlantı kurmak.

Adımlar

1 Mimari Bileşenler

- **Bastion EC2** (Public Subnet, SG: 22/tcp)
- **Database-1 (RDS MySQL)** (Private Subnet, SG: 3306/tcp → sadece Bastion SG'den izinli)



2 Bastion Host Oluřturma

EC2 → Instances → Launch an instance

Name: Bastion Host

[EC2](#) > [Instances](#) > [Launch an instance](#)

Name and tags [Info](#)

Name
 [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

My AMIs

Quick Start


Amazon Linux
aws


macOS


Ubuntu


Windows


Red Hat


SUSE Linux


Debian


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0b09ffb6d8b58ca91 (64-bit (x86), uefi-preferred) / ami-0b2f2a5d7a4d7f37f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description
Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.8.20250908.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	
64-bit (x86)	uefi-preferred	ami-0b09ffb6d8b58ca91	2025-09-06	ec2-user	Verified provider

Instance type: t2.micro

▼ Instance type [Info](#) | [Get advice](#)
Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

☐ All generations [Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Key Pair

- Yeni key pair oluřtur: **ec2_ssh_key**
- Type: **RSA**
- Format: **.pem**
- Dosyayı indirip bilgisayarına kaydet.

Create key pair ✕

Key pair name
Key pairs allow you to connect to your instance securely.

ec2_ssh_key

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** [Learn more](#)

[Cancel](#) [Create key pair](#)

Key dosyasını bilgisayarına kaydet

- **Network settings** → **Edit**:
 - **VPC**: default
 - **Subnet**: varsayılan
 - **Availability Zone**: varsayılan
 - **Auto-assign public IP**: Enable
 - **Firewall (SG)**: Create a new security group
 - **Security group name**: Basiton_SG
 - **Description**: Security group for Basiton_SG
 - **Rules**:
 1. **SSH** → **Source**: Anywhere (0.0.0.0/0)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-02a71ecf125d2118e (default) [↻](#)

Subnet [Info](#)

No preference [↕](#) [↻](#) [Create new subnet](#)

Availability Zone [Info](#)

No preference [↕](#) [↻](#) [Enable additional zones](#)

Auto-assign public IP [Info](#)

Enable [↕](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Security group name - required

Basiton_SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./()#,@!+=&:[]!\$*

Description - required [Info](#)

Security group for Basiton_SG

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0) [Remove](#)

Type Info	Protocol Info	Port range Info
ssh ↕	TCP	22
Source type Info	Source Info	Description - optional Info
Anywhere ↕	Q Add CIDR, prefix list or security group 0.0.0.0/0 X	e.g. SSH for admin desktop

Launch instances butonuna tıkladığınızda, birkaç dakika içinde bastion sunucunuz hazır olacaktır.

Instances (1) [Info](#)

[Q](#) Find Instance by attribute or tag (case-sensitive) [A](#)

☐	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼
☐	Basiton host	i-0bdf2f88cde59a0dc	✔ Running + -	t2.micro

Test için bastion'a bağlan:

```
ssh -i key.pem ec2-user@<BastionPublicIP>
```

```
chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem ec2-user@54.88.146.50
```

```
(base) Downloads ➡ chmod 400 ec2_ssh_key.pem
ssh -i ec2_ssh_key.pem ec2-user@54.88.146.50
The authenticity of host '54.88.146.50 (54.88.146.50)' can't be established.
ED25519 key fingerprint is SHA256:oncqEXEF4zuTxuX6kH6eD+/xMTzRjRZz49GdeHRZPGY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.88.146.50' (ED25519) to the list of known hosts.

#_
~\_ #####_ Amazon Linux 2023
~~ \_#####\
~~ \###|
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~. _ _/
_/_/_/
_/m/'
[ec2-user@ip-172-31-31-17 ~]$
```

3 MySQL Server (RDS) Oluşturma

Aurora and RDS—>Create a Database

[Aurora and RDS](#) > [Dashboard](#)

Resources

Refresh

You are using the following Amazon RDS resources in the US East (N. Virginia) region (used/quota)

DB Instances (0/40)
Allocated storage (0 TB/100 TB)
Instances and storage include Neptune and DocumentDB. [Increase DB instances limit](#)

DB Clusters (0/40)
Reserved instances (0/40)
Snapshots (0)
Manual
DB Cluster (0/100)
DB Instance (0/100)
Automated
DB Cluster (0)
DB Instance (0)
Recent events (0)
Event subscriptions (0/20)

Parameter groups (1)
Default (1)
Custom (0/100)
Option groups (1)
Default (1)
Custom (0/20)
Subnet groups (0/50)
Supported platforms [VPC](#)
Default network vpc-02a71ecf125d2118e

Create a database

Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale a relational database in the cloud.

[Create a database](#)

[Restore from S3](#)

You can use a backup from Amazon S3 to restore and create a new Aurora MySQL and MySQL database.

Note: your DB instances will launch in the **US East (N. Virginia)** region

Service health

View service health dashboard

Current status | Details

Choose a database creation method—>Standard create

Engine options—>MySQL

Create database [Info](#)

Choose a database creation method

☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 

☐ Aurora (PostgreSQL Compatible) 

☒ **MySQL** 

☐ PostgreSQL 

☐ MariaDB 

☐ Oracle 

☐ Microsoft SQL Server 

☐ IBM Db2 

Edition

☒ **MySQL Community**

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

☒ **Show only versions that support the Multi-AZ DB cluster** [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☒ **Show only versions that support the Amazon RDS Optimized Writes** [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Editions: MySQL Community

Templates: Dev/Test

Edition

☒ **MySQL Community**

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

☒ **Show only versions that support the Multi-AZ DB cluster** [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☒ **Show only versions that support the Amazon RDS Optimized Writes** [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.0.42

☐ **Enable RDS Extended Support** [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Templates

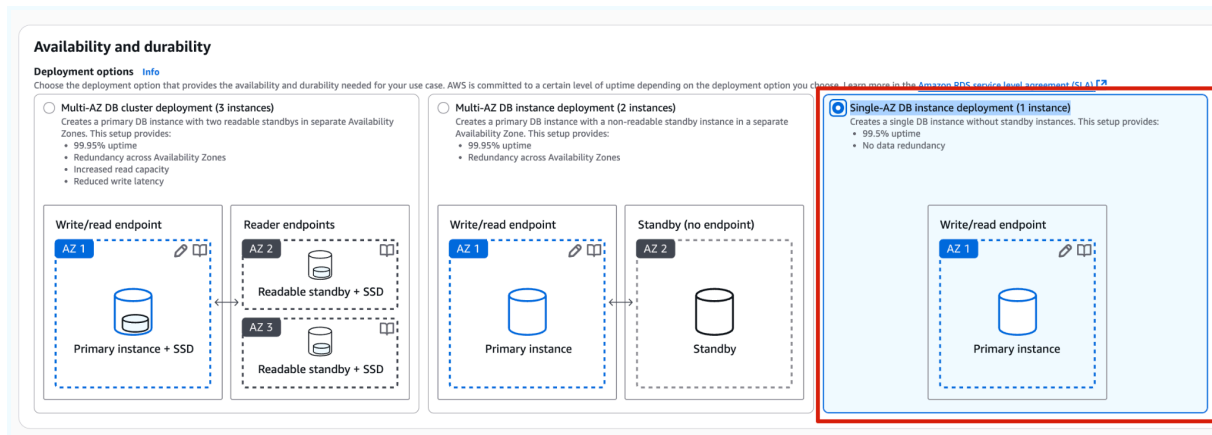
Choose a sample template to meet your use case.

☐ **Production**
Use defaults for high availability and fast, consistent performance.

☒ **Dev/Test**
This instance is intended for development use outside of a production environment.

☐ **Sandbox**
To develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability: Single-AZ DB instance deployment (1 instance)



Settings:

DB instance identifier: **database-1**

Credentials Settings:

Master username: **admin**

Credentials management: **Self managed**

Password: Kendiniz belirleyin.

Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

database-1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ **Credentials Settings**

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ **Managed in AWS Secrets Manager - most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ **Auto generate password**
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength **Strong**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

Instance configuration:

DB instance class→Burstable classes (includes t classes)

Instance type: **db.t3.micro**

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | [Info](#)

▼ **Hide filters**

☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

db.t3.micro
2 vCPUs 1 GiB RAM Network: Up to 2,085 Mbps

Storage

Storage type: **default**

Allocated storage: **20**

Storage

Storage type | [Info](#)
Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp3)
Performance scales independently from storage

Allocated storage | [Info](#)
20 GiB
Minimum: 20 GiB. Maximum: 6,144 GiB

Provisioned IOPS | [Info](#)
3,000 IOPS
Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GiB.

Storage throughput | [Info](#)
125 MiBps
Baseline storage throughput of 125 MiBps is included for allocated storage less than 400 GiB.

ⓘ To provision additional IOPS and throughput, increase the allocated storage to 400 GiB or greater.

► **Additional storage configuration**

connectivity→Connect to an EC2 compute resource

EC2 instance: **Bastion host**

Network type: **IPv4**

Virtual private cloud (VPC): **default**

Connectivity [Info](#)

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

EC2 instance [Info](#)
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0bdf2f88cde59a0dc
Bastion host

Some VPC settings can't be changed when a compute resource is added
Adding an EC2 compute resource automatically selects the VPC, DB subnet group, and public access settings for this database. To allow the EC2 instance to access the database, a VPC security group rds-ec2-X is added to the database and another called ec2-rds-X to the EC2 instance. You can remove the new security group for the database only by removing the compute resource.

Network type [Info](#)
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-02a71ecf125d2118e)
6 Subnets, 6 Availability Zones

Only VPCs with corresponding DB subnet groups are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☐ Choose existing
Choose existing DB subnet group

☒ **Automatic setup**
RDS creates a new subnet group for you or reuses an existing subnet group

DB subnet group: Automatic setup
Public access: No
VPC security group (firewall)-->Create new
New VPC security group name: Mysql

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

☐ Choose existing
Choose existing DB subnet group

☒ **Automatic setup**
RDS creates a new subnet group for you or reuses an existing subnet group

DB subnet group name
rds-ec2-db-subnet-group-1
New DB subnet group created.

Public access [Info](#)
☐ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database.

☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing
Choose existing VPC security groups

☒ **Create new**
Create new VPC security group

New VPC security group name
Mysql

Amazon RDS will add a new VPC security group rds-ec2-1 to allow connectivity with your compute resource.

Availability Zone [Info](#)
us-east-1c

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

Diğer tüm ayarları varsayılan olarak kabul edip **Create database** tıklayın

👉 Database Status "Available" olana kadar bekleyin.

Creating database database-1

Your database might take a few minutes to launch. You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

Databases (1)

Group resources

Modify

Filter by databases

DB identifier	Status	Role	Engine	Region ...	Size	Recommen
database-1	Creating	Instance	MySQL Co...	us-east-1c	db.t3.micro	

Aws console'da database-1 'e tıklayarak database'e bağlanabilmemiz için :

👉 RDS endpoint adresini not edin.

Aurora and RDS

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved Instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL Integrations

Events

Event subscriptions

Recommendations

Certificate update

database-1

Summary

DB identifier database-1

CPU 6.75%

Status Available

Class db.t3.micro

Role Instance

Current activity 0 Connections

Engine MySQL Community

Region & AZ us-east-1c

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL Integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Endpoint port database-1.c4t40ua0c47v.us-east-1.rds.amazonaws.com 3306

Availability Zone us-east-1c

VPC vpc-02a71ec125d2118e

Subnet group rds-ec2-db-subnet-group-1

Subnets subnet-06fa55cb00ea6318b subnet-012b16af30a0749f4 subnet-0d9050b17a4def033 subnet-0f69500f89a2853a9 subnet-0fef10978ed09e9c4

Network type IPv4

Security VPC security groups MySQL (sg-00be31d168973ba72) rds-ec2-1 (sg-0b64ca3a19ca8c6c8) Publicly accessible No Certificate authority rds-ca-rsa2048-g1 Certificate authority date May 26, 2061, 02:34 (UTC+03:00) DB instance certificate expiration date September 15, 2026, 11:36 (UTC+03:00)

4 SSH Tunnel ile MySQL'e Bağlantı

DBeaver(<https://dbeaver.io/download/>) veya mysql workbench(<https://dev.mysql.com/downloads/workbench/>) gibi programları kullanarak Mysql'e ssh tunnel ile bağlanabiliriz.

◆ DBeaver üzerinden:

Connection Settings→SSH

Host/IP: Bastion IP Public

User Name: ec2-user

Authentication method: Public key

Add bastion host ssh key:ec2_ssh_key.pem

Test tunnel connection butonuna basarak bağlantınızı test edin.

Connect to a database

MySQL connection settings

Main | Advanced | Driver properties | SSH X

Settings

Host/IP: 54.88.146.50 Port: 22

User Name: ec2-user

Authentication Method: Public Key

Private Key: /Users/hakan/Downloads/ec2_ssh_key.pem

Passphrase

Save credentials

Jump servers

Advanced settings

Test tunnel configuration

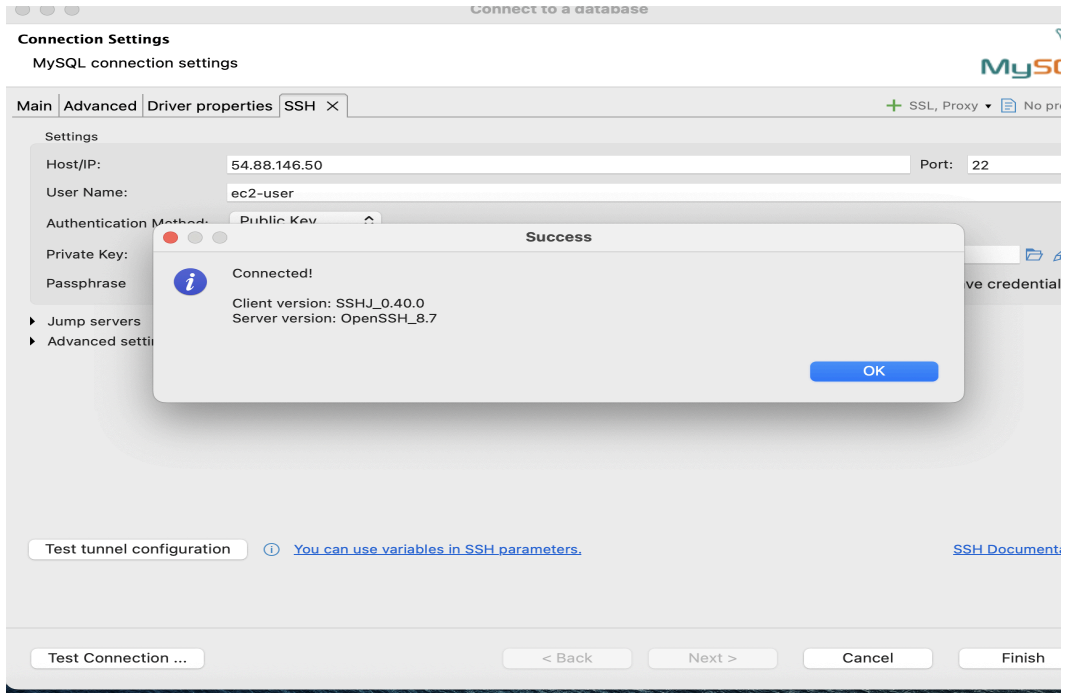
You can use variables in SSH parameters.

SSH Documentation

Test Connection ...

< Back Next > Cancel Finish

Bastion Host sunucusundan RDS Mysql'e başarılı şekilde SSH tunnel kuruldu.



Şimdi RDS mysql connection ayarlarını yapalım.

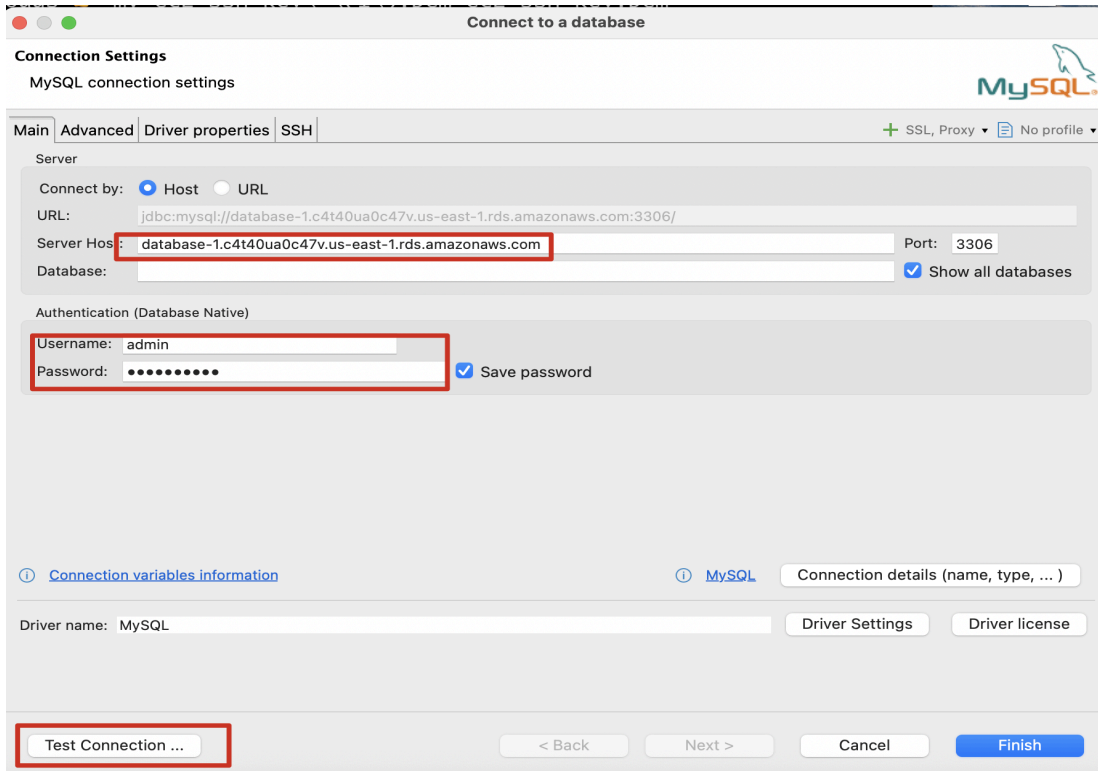
Connection Settings→**Main**

Server Host: RDS Endpoint

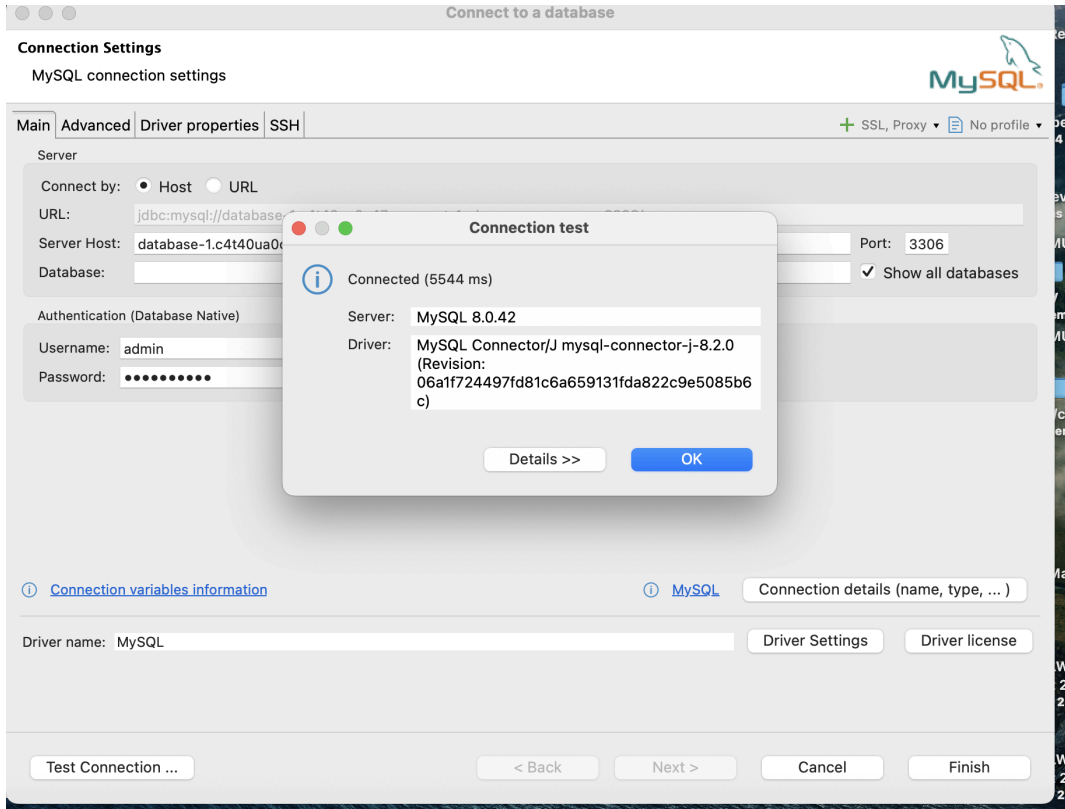
Username: Admin

Password: Şifreniz..

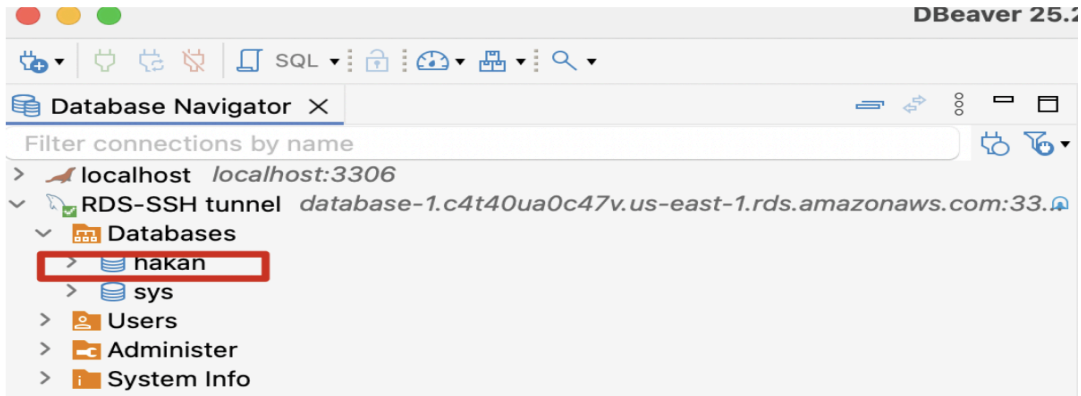
Test Connection butonuna basarak RDS mysql'e bağlanabiliriz.



Bağlantı başarılı



Hakan adında **database** oluşturduk.



♦ Terminal üzerinden:

SSH Tunnel (lokal port forward):

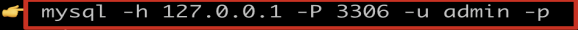
Yerel makineden SSH Tunnel kur:

```
ssh -i ec2_ssh_key.pem -L 3306:<PrivateDBIP>:3306  
ec2-user@<BastionPublicIP>
```

```
ssh -i ec2_ssh_key.pem -L  
3306:database-1.c4t40ua0c47v.us-east-1.rds.amazonaws.com:3306  
ec2-user@54.88.146.50
```

MySQL'e bağlan:

```
mysql -h 127.0.0.1 -P 3306 -u admin -p
```

```
ec2-user@ip-172-31-31-17:~ (ssh) 361 X mysql 362  
Last login: Mon Sep 15 11:07:57 on ttys000  
You have mail.  
(base) ~  mysql -h 127.0.0.1 -P 3306 -u admin -p  
Enter password:  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 37  
Server version: 8.0.42 Source distribution  
  
Copyright (c) 2000, 2018, Oracle and/or its affiliates. All rights reserved.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
mysql> show databases;  
+-----+  
| Database |  
+-----+  
| hakan    |  
| information_schema |  
| mysql    |  
| performance_schema |  
| sys      |  
+-----+  
5 rows in set (0.17 sec)
```

Cleanup

Lab tamamlandıktan sonra maliyet çıkmaması için:

- Bastion Host → **Terminate**
- RDS MySQL → **Delete**